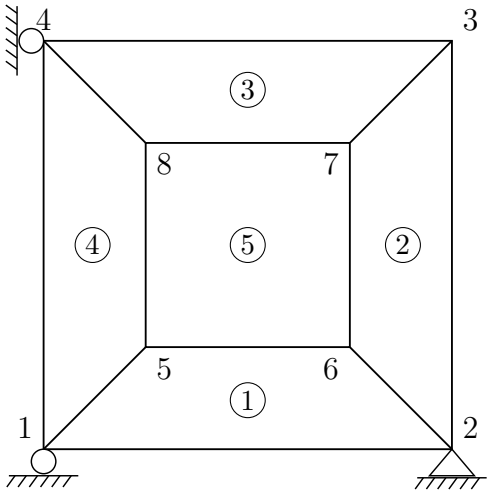
P1. **[10 points]**

The finite element mesh for a two-dimensional elastostatic boundary-value problem is shown in the figure below. Set up the ID, IEN and LM arrays for the following mesh of four-node quadrilateral elements. Assume the node with the lowest global number is number one in the local ordering. Number the remaining local nodes counterclockwise.



P2. **[10 points]**

Let

$$W_c = -\frac{\nu}{2E}(\sigma_{kk})^2 + \frac{1+\nu}{2E}\sigma_{ij}\sigma_{ij}$$

denote the complementary energy, where ν = Poisson's ratio and E = Young's modulus of elasticity. Using the indicial notation, show that

$$\frac{\partial W_c}{\partial \sigma_{ij}} = -\frac{\nu}{E}\sigma_{kk}\delta_{ij} + \frac{1+\nu}{E}\sigma_{ij}.$$

Caution: This problem will be graded on the basis of how you used the indicial notation, and not so much about being able to get the derivatives, so be careful with your use of the indices.

P3. **[10 points]**

The state of stress in a body is given by $\sigma_{xx} = \sigma_1$, $\sigma_{yy} = \sigma_2$, $\sigma_{zz} = \sigma_3$, and $\sigma_{xy} = \sigma_{yz} = \sigma_{zx} = 0$. Consider a plane whose unit normal has the same direction cosine with respect to the positive x, y, and z axes. Show that on this plane the normal component of traction σ and resolved shear component of traction τ are given by

$$\sigma = \frac{1}{3}(\sigma_1 + \sigma_2 + \sigma_3)$$

and

$$\tau = \frac{1}{3}\sqrt{(\sigma_1 - \sigma_2)^2 + (\sigma_2 - \sigma_3)^2 + (\sigma_1 - \sigma_3)^2}.$$

P4. **[10 points]**

Consider a CST on the (x, y)-plane with the following element nodes: Node #1 with coordinates $(0, 0)$, Node #2 at $(2, 1)$, and Node #3 at $(1, 2)$. It is desired to construct the vector of strains for element e under the plane strain condition with $\epsilon_{zz} = 0$:

$$\begin{Bmatrix} \epsilon_{xx} \\ \epsilon_{yy} \\ \gamma_{xy} \end{Bmatrix}^e = \mathbf{B}\mathbf{d}^e,$$

where γ_{xy} is the engineering shear strain and $\mathbf{d}^e$ is the element displacement vector. Derive the local element shape functions in terms of x and y for this triangle, i.e., $N_a(x, y)$ for $a = 1, 2, 3$, and define the matrix $\mathbf{B}$ above, i.e., write out all elements of this matrix.

Use the displacement ordering

$$\mathbf{d}^e = \begin{bmatrix} u_{x1} & u_{y1} & u_{x2} & u_{y2} & u_{x3} & u_{y3} \end{bmatrix}^\mathsf{T}.$$

P5. **[10 points]**

Consider the same CST element of Question P4. It is desired to augment the vector of strains with rotation, so that the augmented vector is now given by

$$\left\{ \begin{array}{c} \epsilon_{xx} \\ \epsilon_{yy} \\ \epsilon_{xy} \\ \omega_{xy} \end{array} \right\}^e = \widetilde{\mathbf{B}}\mathbf{d}^e,$$

where ϵ_{xy} is the tensorial shear strain and $\omega_{xy} = (u_{x,y} - u_{y,x})/2$ is the xy component of the infinitesimal rotation tensor. Write out all of the elements of the matrix $\widetilde{\mathbf{B}}$.

P6. **[10 points]**

Consider the prototype problem $u_{,xx} + f = 0$ subject to boundary conditions. Assume a mesh is created so that the element length is constant (i.e., $\Delta x_A = \Delta x$, $A = 1, 2, \ldots, n$). Consider the finite difference "stencil" at a typical internal node, namely,

$$\frac{u_{A+1} - 2u_A + u_{A-1}}{\Delta x^2} + f_A = 0.$$

Take $f(x) = x^2$. Using piecewise linear shape functions, set up the finite element equation associated with node A and contrast it with the finite difference equation given above assuming $x_{A-1} = 0$, $x_A = 1$, and $x_{A+1} = 2$.

P7. **[10 points]**

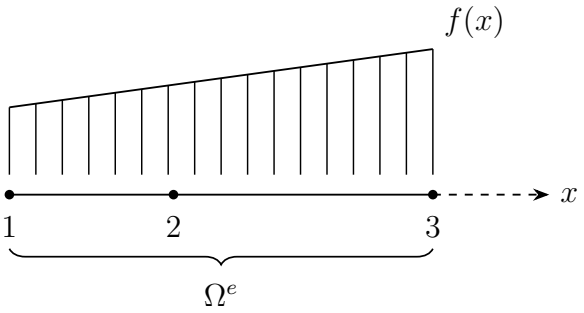
A three-node quadratic element is shown in the figure below. Assume the coordinates of local nodes 1, 2 and 3 are given as x_1^e, x_2^e, and x_3^e, respectively. Use isoparametric interpolation to establish a general mapping $x = x(\xi)$ in Ω^e and determine the lowest-order Gaussian integration formula (i.e., $n_{\text{int}} =?$) which exactly integrates

$$F_a^e = \int_{\Omega^e} N_a f \, \mathrm{d}x$$

if:

(a) f is linear in x and $x_2^e \neq (x_1^e + x_3^e)/2$;

(b) f is linear in x and $x_2^e = (x_1^e + x_3^e)/2$.



P8. **[10 points]**

Consider a three-node element in one dimension in which the monomials for the shape functions are to be 1, ξ, and ξ^m, where m is a nonnegative integer. The local coordinates of the nodes are $\xi = -1$, 0 and 1. For which values of m can the shape-function coefficients be determined uniquely?

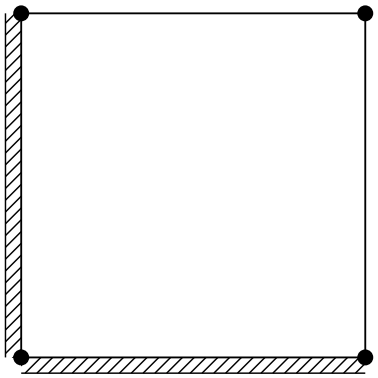
P9. **[10 points]**

Determine the constraint ratio r for a 4-node quadrilateral element (see the figure below) integrated using:

(a) a one-point rule;

(b) a 2×2 integration rule;

(c) a 3×3 integration rule.



P10. **[10 points]**

Determine the rank of the element stiffness matrix $\mathbf{k}^e$ for the following elements:

(a) three-node triangle for fluid conduction;

(b) four-node quadrilateral element for fluid conduction;

(c) CST for solid mechanics;

(d) four-node quadrilateral for solid mechanics;

(e) eight-node hexahedral element for solid mechanics.

Assume standard (full) numerical integration is used for all elements.